

Facilities Design: Group Project brief (manufacturing 1 finished product, complex production processes involving machines and manual work)

Hypnos Mattress Manufacturers is a leading South African producer of premium-quality mattresses. A recent increase in market demand left the company unable to fulfil production requirements with their current strategy. The board of directors decided to develop a new facility specifically for the company's double-size pocket spring mattress, thereby freeing up production capacity for their other product lines in the current facility. At the new facility the production processes have to be redesigned to meet the increased demand. Current demand projections are estimated at 200 000 mattresses per year. The management team appointed a facility design team to develop a detailed facility plan to aid with the strategic decision of where the *new* facility should be located.

To maintain their competitive edge and signature mattress style, Hypnos Mattress Manufacturers produces its own polyurethane *foam* with a unique chemical structure for the internal cushioning layers on-site, rather than relying on external suppliers for this key component of their mattresses. The core of the pocket spring mattress consists of *wire springs* wrapped in a lightweight non-woven polypropylene fabric, which is secured in place through ultrasonic or thermal welding of the fabric. The additional cushioning and protective layers of the mattress are then added and sewn into place before packaging the mattresses for distribution to customers. The company's current production process can be seen in the following video:

<https://www.youtube.com/watch?v=S2UjKJCQlxc>

The design team must use the production process illustrated in this video as a base for the production process design used in the new facility, with adjustments and upgraded material handling methods as required to develop the new facility.

Product specifications

The dimensions and internal structure of the mattress are shown in Figure 1.

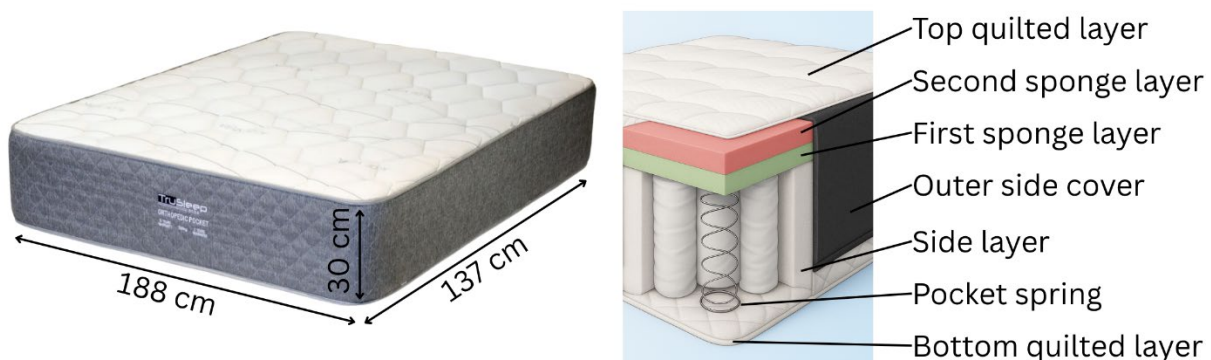


Figure 1: Double Pocket Spring Mattress Dimensions and Internal Structure

The company's current raw material suppliers are provided in Table 1. As the new facility will be developed as a separate entity from the existing production plant, no supplier contracts are in place and any of these suppliers could be replaced if the design team deems alternative suppliers more suitable.

Table 1: Raw Materials and Current Suppliers

Mattress Component	Application	Material	Current Supplier
Polyurethane foam	Sponge cushion layer, two hardnesses per mattress.	Liquid polyol (have to be stored outside the building for safety reasons)	https://gzsunray.en.made-in-china.com/product/tFUGClmTJNcf/China-Polyurethane-PU-Flexible-Mattress-Foam-Systems-Blend-Polyol-Isocyanate-with-Good-Price.html
Pocket spring cover fabric	Cover and connect the wire springs.	White 60 gsm non-woven polypropylene	https://vitatex.co.za/vitatex-products/spunbond/spunbond-technical-datasheet/
Wire springs	Main mattress structure.	1.8 mm diameter steel wire coils	https://www.wiresupplies.co.za/products/bending-seating-wire/
Edge rod	Stiffness around the mattress edge.	3 mm diameter steel wire	Same supplier as spring wire coil
Internal side cover	Insulation and protective layer around the side of the springs.	Needle punch felt	https://www.ginnispectra.com/needle-punch-felt.html *
Mattress outer top	Outer quilted mattress layer.	Premium ticking fabric	https://sinocomfort.com/product/soft-and-breathable-mattress-ticking-fabric/ *
Outer quilt batting	Layer for additional plushness and quilting aesthetics.	Dacron polyester batting	https://vnpolyfiber.com/product/dacron-polyester-batting-in-upholstery/ *
Side cover	Durable outer side cover.	Woven ticking fabric	https://ladetech.en.made-in-china.com/product/ywFfUdgJflro/China-High-Quality-Mattress-Jacquard-Ticking-Knitted-Fabric-180-450-GSM.html *

*Make own assumptions on raw material dimensions where not provided.

Consumables include sewing thread, branding labels, staples for attaching the edge rod to the pocket spring unit, packaging materials and machine spares. No suppliers need to be considered for these items, as the company has existing contracts with suppliers. Storage and material handling requirements for these items must still be considered in the design.

Operation specifications

Although the process is already defined below, management expects you to recommend appropriate equipment as part of the production process design. Present the process in a detailed process flow chart.

Table 2: Mattress manufacturing process steps and estimated processing times

Production Step	Estimated Processing Time
Polyurethane <i>foam production</i> : piping, expansion, setting, demoulding and plastic wrapping.	12 min per large slab
Foam <i>pressing</i>	2 min per large slab
Foam <i>cutting</i> . Assume each foam slab produces 8 <i>sheets</i> of foam	8 min per large slab (1 min per sheet)
Wire springs <i>coiling</i>	<i>Show own calculations</i>
Spring <i>wrapping</i> .	<i>As above</i>
<i>Arrange, cut and align</i> springs to form mattress base	<i>As above</i>
<i>Seal</i> top and bottom cover sheets	5 min per mattress
Automated <i>quilting</i> (two layers of ticking fabric with batting in between).	30 s per meter of quilt produced
Hand <i>finishing</i> of quilt.	1 m per quilt
<i>Sewing</i> of side cover: wide strip, narrow strip and logo.	2 min per set
Side cover <i>assembly</i> .	12 min per mattress
Mattress <i>assembly</i> : add and sew 2 foam sheet layers, top quilt and bottom quilt.	15 min per mattress
Preparation for packing include few steps: plastic <i>wrap</i> , <i>press</i> , <i>vacuum seal</i> and roll mattress.	13 min per mattress
Final <i>quality check</i> , <i>box/pack</i> and <i>seal</i> .	12 min per mattress

Assume that setup times is included in process times.

Recommended machinery for all critical manufacturing processes must be provided.

All assumptions regarding lot sizes, equipment used and process speed should be mentioned and justified accordingly. Machinery equipment specifications are critical to consider and must be clearly shown in calculations.

Scrap is only produced where the foam part of the mattress is produced. Assume 5% scrap and propose scrap handling and management strategies. Assume no scrap at springs production, as the machines are highly efficient and calibrated daily.

Manufacturing and management policies to be considered when designing facilities:**1. Manufacturing Operating Times:**

The company operates for one 9-hour shift per day, 5 days a week. Operators receive a 30-minute lunch break and two 15-minute tea breaks per day. A total of 220 working days per year can be assumed. Overtime or a Saturday shift is possible, but management would prefer to avoid it where possible, unless flexible capacity is required for unforeseen spikes in demand.

2. Inventory Levels:

The company's inventory policy stipulates that 5 days' worth of raw materials must be kept in stock due to long supplier lead times. Finished goods storage must be sufficient for 2 weeks of stock, as the Hypnos Mattress Manufacturers has an international customer footprint and needs to ship large batches at a time. The Work-in-Progress inventory levels and space allocation must be determined by the design team.

3. Deliveries and Collections:

The facility must allow space for small order collection on-site by customers and the dispatch of large orders from the facility. Hypnos Mattress Manufacturers has its own fleet of delivery trucks serving the current facility, but the fleet will not have sufficient capacity to transport the new facility's products as well. The design team must advise the management on a suitable *delivery protocol for bulk orders* from the new facility. Also advise management on a raw material delivery schedule dependent on the facility warehousing capacity. The finished goods dispatch schedule must also be recommended based on the design of the facility and the delivery protocol.

Management policies:

- Raw material *receival* and *storage/warehouse* areas, material and component *staging* areas during manufacturing and assembly, as well as Finished goods *storage/warehouse* and *staging* areas must be clearly indicated.
- Clear indication on *entry and exit* to the premises must be visible on the layout.
- *Parking* must be provided for all personnel. Personnel include management, administration and operators. Reasoning and calculations for determining parking space must be provided.
- *Lockers* for plant personnel are provided as well as a cafeteria providing basic food, but no Health services are provided. There should be enough Rest Rooms provided in both offices as well as in the Manufacturing areas, as per standards.
- Maintenance and Spare parts *storage rooms* need to be available in the facility, although the layout detail of those do not have to be done.
- *Office space* must be allocated to the Senior Plant Manager, Operations Manager, Inventory Manager, the administrative team, including Reception (for customers and suppliers visiting the offices), Purchasing, Sales, Finance, Marketing and Human Resources employees, and the plant

supervisors. The number of supervisors will depend on the facility design and must be clarified by the design team.

- Make logical assumptions regarding Facility support systems eg. Lightning, Atmospheric, Safety and other factory requirements. Management is considering sustainable energy options. Recommend at least one other strategy they can consider incorporating into the new facility design.
- Utility considerations, including electricity, water and plumbing should be discussed, even though it is not necessary to indicate in the layout.
- As Hypnos mattresses are very popular, management expects 15% extra space allowed for expansion possibilities in the next 5 years.
- Define small and large orders that makes sense depending on warehouse policy and dispatch schedules.